Profitability & Cost-to-Serve Analysis

1. Executive Summary

Analyzed a synthetic dataset of 227,000 orders across 10,000 customers to assess customer-level profitability and cost-to-serve. 227,243 completed orders, 9,998 active customers, $3.58B net revenue.

After properly accounting for product cost, the business earns a **True Contribution Margin of approximately 26.4%**, providing a more realistic view of profitability. Operational Cost-to-Serve — including shipping, storage, customer service, and returns — represents approximately **0.46% of revenue**.

Found customer service and shipping accounted for 88% of cost-to-serve, highlighting opportunities for cost optimization and pricing reviews. Together they are the **largest Cost-to-Serve drivers**.

Profit is meaningfully concentrated across the customer base: approximately half of all customers generate 80% of true contribution profit.

Calculated a 26.4% contribution margin and identified **13 loss-making customers** requiring profitability reviews. All 13 sit within the SMB segment.

A material correction was made during the project review. The original Contribution Profit metric did not subtract Product Cost (COGS), producing an unrealistic 99.5% margin. The profitability logic was reviewed and corrected, and this report presents the corrected profitability results.

2. Business Problem

Aggregate revenue growth can mask uneven customer-level economics.

Some customers may be highly profitable, while others may consume disproportionate service and fulfillment costs relative to the revenue they generate.

Without customer-level Cost-to-Serve and profitability visibility, management cannot make informed decisions about pricing, service investment, customer retention, or account prioritization.

3. Objectives

- Quantify true profitability at the business and customer level.
- Identify the main Cost-to-Serve drivers.
- Analyze customer profitability and cost efficiency.
- Measure profit concentration through Pareto analysis.
- Identify genuinely loss-making customers.
- Translate analytical findings into specific business recommendations.

4. Data Overview

Entity	Records
Orders (all statuses)	250,000
Completed orders (analytical population)	227,243
Order line items	512,515
Customers	10,000
Customers with ≥1 completed order	9,998
Products	1,000
Regions	30
Warehouses	20

Order Status Distribution

- Completed: 227,243
- Returned: 12,796
- Cancelled: 9,961

Only completed orders are included in the primary profitability analysis, consistent with the project's business rules.

5. Architecture

The project follows an end-to-end analytical workflow:

```
10 Source CSVs
      ↓
SQL Server Staging
      ↓
Data Quality Audit
      ↓
Validated Core Tables
      ↓
Reporting & Analytical Views
      ↓
Power BI Data Model
      ↓
Executive Dashboard
      ↓
Business Insights & Recommendations
```

The SQL layer includes a **32-check automated data-quality audit** covering key validation areas such as uniqueness, referential integrity, financial validity, date validity, and business rules.

6. Data Quality

The dataset was reviewed through automated validation checks covering:

- Uniqueness
- Referential integrity
- Financial validity
- Date validity
- Business-rule validation
- Transactional reconciliation

The dataset was found to be internally consistent, including:

- Zero orphaned foreign-key records
- No negative-cost violations
- Successful reconciliation of core transactional loads

A load-logic issue involving Customer Service records without a matching order was identified during review and corrected so that such records are captured through the audit process rather than silently dropped.

7. Methodology

Revenue and Product Cost are calculated at the order-line level and then aggregated to order and customer levels.

Cost-to-Serve combines four major cost categories:

- Shipping
- Customer Service
- Storage
- Returns

Gross Profit is calculated after subtracting Product Cost (COGS) from Revenue.

True Contribution Profit is then calculated after subtracting Cost-to-Serve from Gross Profit.

Only completed orders are included in the primary profitability analysis.

Storage cost is allocated based on units sold within the same warehouse, product, and month.

8. KPI Overview

KPI	Value
Net Revenue	$3,576,677,378
Product Cost (COGS)	$2,615,275,667
Gross Profit	$961,401,712

KPI	Value
Gross Margin	26.88%
Total Cost-to-Serve	$16,523,929
Cost-to-Serve Rate	0.46%
True Contribution Profit	**$944,877,783**
True Contribution Margin	**26.42%**

9. Profitability Insights

True Contribution Margin is **26.42%**, compared with a Gross Margin of **26.88%**.

This indicates that Cost-to-Serve creates a real but relatively modest reduction in overall profitability.

The corrected profitability view is materially different from the original 99.5% margin, which resulted from excluding Product Cost from the Contribution Profit calculation.

All three customer segments generate relatively similar true margins:

- Enterprise: 25.66%
- Mid-Market: 26.67%
- SMB: 27.36%

This indicates that overall profitability differences are driven more by customer-level economics and Cost-to-Serve efficiency than by broad segment-level margin differences.

10. Cost-to-Serve Insights

Driver	Cost	Share of Total Cost-to-Serve
Customer Service	$9,696,715	58.7%
Shipping	$4,847,989	29.3%
Storage	$1,199,899	7.3%
Returns	$779,325	4.7%
Total	**$16,523,929**	**100%**

Customer Service is the dominant Cost-to-Serve driver, accounting for **58.7%** of total Cost-to-Serve.

Shipping is the second-largest driver at **29.3%**.

Found customer service and shipping accounted for **88% of cost-to-serve**, highlighting opportunities for cost optimization and pricing reviews.

11. Customer Profitability

Segment	Customers	Revenue	Gross Profit	True Contribution Profit	True Margin
Enterprise	1,503	$1,462.3M	$379.8M	$375.3M	25.66%
Mid-Market	3,516	$1,285.9M	$349.0M	$342.9M	26.67%
SMB	4,979	$828.4M	$232.6M	$226.6M	27.36%

Enterprise customers generate the largest absolute contribution profit, primarily because of their higher revenue and order volume rather than significantly higher margins.

The segment-level margins remain relatively close to one another, indicating that customer-level profitability and cost efficiency are more important analytical dimensions than segment membership alone.

12. Customer Risk

13 customers are genuinely loss-making based on True Contribution Profit, and all 13 belong to the SMB segment.

The most significant example is **Meridian Holdings Co. 2**, which loses approximately $80K on approximately $99K of revenue.

Its Cost-to-Serve is approximately $108K compared with Gross Profit of approximately $28K, making it a strong candidate for pricing, service-model, or shipping-cost review.

The corrected profitability methodology therefore provides a more realistic view of customer risk than the original profitability metric.

13. Pareto / Profit Concentration

Approximately **50% of customers (5,042 of 9,998)** account for approximately **80% of true contribution profit**.

This represents moderate profit concentration rather than an extreme 80/20 dependency.

The result indicates that profitability is relatively broad-based across the customer portfolio, reducing dependence on a very small number of accounts.

14. Business Drivers

Revenue is broadly diversified across the seven industries represented in the dataset:

- Retail
- Professional Services
- Education
- Healthcare
- Finance

- Technology
- Manufacturing

Industry-level true margins are relatively consistent, ranging from approximately **26.3% to 26.5%**.

Revenue is also relatively balanced across industries, ranging from approximately **$484M to $531M**.

This suggests that the primary profitability opportunity is not industry mix, but rather **customer-level Cost-to-Serve efficiency and account economics**.

15. Key Findings

1. Calculated a **26.4% contribution margin** — materially different from the original 99.5% metric.
2. Customer Service is the largest Cost-to-Serve driver, representing **58.7%** of total Cost-to-Serve.
3. Found customer service and shipping accounted for **88% of cost-to-serve**.
4. **13 loss-making customers** requiring profitability reviews — all SMB.
5. Approximately **50% of customers generate 80% of true contribution profit**, indicating moderate profit concentration.
6. Segment and industry margins are relatively uniform, making customer-level economics and Cost-to-Serve efficiency more important profitability drivers.

16. Business Recommendations

1. **Investigate the 13 loss-making SMB customers** individually to determine whether pricing, service levels, minimum-order policies, or shipping methods are creating structural losses.

2. **Prioritize Customer Service cost efficiency**, as Customer Service is the largest Cost-to-Serve driver.

3. **Review shipping methods and fulfillment decisions** for cost-sensitive customers and segments.

4. **Protect and grow the highest-value customers** responsible for the majority of contribution profit through retention-focused account management.

5. **Continue validating profitability metrics and downstream customer analyses** whenever the underlying profitability methodology changes.

17. Management Actions

Action	Owner	Priority
Apply the corrected True Contribution Profit logic to the Power BI report	BI / Analytics Team	Immediate
Review 13 loss-making SMB accounts	Account Management	High
Launch Customer Service cost-efficiency initiatives	Operations	High

Action	Owner	Priority
Review shipping methods for cost-sensitive accounts	Operations / Supply Chain	High
Develop retention actions for top profit-driving customers	Sales / Account Management	Medium
Re-validate downstream profitability analyses after metric changes	Analytics Team	Medium

18. Future Opportunities

- Re-export the Power BI dashboard screenshots once the report visuals are re-pointed at the corrected True Contribution Profit metrics, since the SQL views, dashboards and analysis are done.
- Extend the Power BI dashboard with product-level and region-level profitability analysis.
- Investigate the root causes of the 13 loss-making accounts at order level to distinguish one-off outliers from structural pricing or service issues.
- Add additional performance optimization, such as indexing and query tuning, if the dataset is significantly expanded.
- Introduce a lightweight orchestration layer, such as SQL Server Agent or Azure Data Factory, if the project evolves into a recurring production data pipeline.

19. Conclusion

This project demonstrates how transactional enterprise data can be transformed into actionable profitability intelligence through SQL Server and Power BI.

The analysis goes beyond revenue reporting by combining Product Cost, Cost-to-Serve, customer profitability, risk identification, and profit concentration analysis.

The most important analytical lesson from the project review was the need to validate business metrics before drawing conclusions. Correcting the Contribution Profit calculation changed the profitability picture from an unrealistic 99.5% margin to a more credible **26.42% True Contribution Margin** and revealed **13 genuinely loss-making customers** that required further investigation.

The resulting analysis provides management with a clearer view of where profit is generated, where operational costs are concentrated, and which customer relationships deserve closer attention.